Complete manuscript comparison

Overview v12: symmetric manuscript counts

This additive edition restores three whole-topic comparisons and the three affected individual comparisons after exact executable-listing exclusions on both sides. It retains 19 topics and 43 recovered originals among 106 accessible originals; 63 remain pending. Scientific admissions, required proof inputs and the 55 archived originals are unchanged. Expanded E and enlarged Mixes are outside this edition.

The complete written factor is $\sum_i c(S_i)/(c(U) + \sum_i c(A_i))$ within each topic: the whole common paper and every complete application are paid. No theory is free because another topic proves a similar statement. The primary unit is characters without whitespace; each topic retains its published source representation and extraction convention. These are observed manuscript costs, not minimum proof complexities. No pooled factor is formed.

Topic	Originals	Sources	Common	Applications	Factor
A	7	213,803	49,059	87,019	1.571
B	5	124,906	44,905	69,265	1.094
C	5	118,433	30,437	61,248	1.292
D	3	47,894	15,632	29,855	1.053
E	3	71,375	19,790	19,513	1.816
F	2	69,825	15,876	27,398	1.614
G	2	73,199	30,928	85,207	0.630
Affine	1	32,767	19,070	22,638	0.786
Representation	1	25,348	11,288	13,688	1.015
Chamber	2	34,177	15,199	21,254	0.938
Matching	1	37,951	16,732	17,168	1.119
ChiralEnumeration	1	27,594	8,831	23,171	0.862
Operations	3	200,630	58,215	100,379	1.265
Gluing	1	27,081	28,416	11,565	0.677
RankContraction	1	15,530	6,937	10,555	0.888
LocallyProjective	2	26,490	25,498	11,966	0.707
MathieuCensus	1	13,736	5,516	35,862	0.332
RankAugmentation	1	33,604	13,904	14,096	1.200
Mixes	1	57,753	12,615	31,077	1.322

The exact 19-row topic CSV retains historical costs and fractions alongside current values. Chamber, ChiralEnumeration and Operations now use the matched listing boundary. The other 16 topic records are unchanged. A factor below one remains visible.

Individual comparisons: first 22 originals

After acquiring the *whole* common theory and all required proof, program and data premises for that topic, the conditional factor is $c(S_i)/c(A_i)$. Every complete local application or companion bundle is charged. Optional validation is not relabeled as a premise.

Original	Topic	Source	Application	Factor
L032	A	16,558	8,212	2.016
L034	A	21,433	5,784	3.706
F016	A	43,983	15,419	2.853
F013	A	25,562	6,224	4.107
L035	A	51,019	14,498	3.519
N009	A	36,824	28,114	1.310
L128	A	18,424	8,768	2.101
N001	B	14,519	7,371	1.970
N003	B	17,780	9,155	1.942
N008	B	18,009	11,132	1.618
L030	B	22,704	12,286	1.848
L040	B	51,894	29,321	1.770
L088	C	13,166	11,311	1.164
N013	C	27,361	15,815	1.730
L108	C	12,849	10,075	1.275
F014	C	20,577	6,962	2.956
L068	C	44,480	17,085	2.603
L066	D	20,916	10,989	1.903
L074	D	20,661	13,367	1.546
L104	D	6,317	5,499	1.149
L083	E	20,885	3,002	6.957
L079	E	20,908	1,855	11.271

Only L008, L028 and N006 change. Their source listings were previously included while replacement code was excluded. Current source costs remove the exact reviewed blocks; application costs remain unchanged. The other 40 conditional fractions retain their earlier values, including N016, N005 and N007. All 43 rows, exact fractions and acquisition conditions are in applications.csv.

Programs implementing proofs keep that logical role even when their listing text is excluded symmetrically from a manuscript comparison. The complete program and mandatory-data inventories remain separately bound. There is no software-inclusive or optional-validation-inclusive scalar: historical source software costs are unknown.

Individual comparisons: remaining 21 originals

Original	Topic	Source	Application	Factor
L021	E	29,582	14,656	2.018
L014	F	25,574	11,504	2.223
L007	F	44,251	15,894	2.784
L009	G	49,891	23,544	2.119
L092	G	23,308	61,663	0.378
L012	Affine	32,767	22,638	1.447
F010	Representation	25,348	13,688	1.852
L008	Chamber	20,159	7,571	2.663
N016	Chamber	14,018	13,683	1.024
F015	Matching	37,951	17,168	2.211
L028	ChiralEnumeration	27,594	23,171	1.191
N005	Operations	122,002	49,595	2.460
N006	Operations	55,740	38,479	1.449
N007	Operations	22,888	12,305	1.860
L010	Gluing	27,081	11,565	2.342
L037	RankContraction	15,530	10,555	1.471
L084	LocallyProjective	2,071	4,857	0.426
L095	LocallyProjective	24,419	7,109	3.435
L073	MathieuCensus	13,736	35,862	0.383
L020	RankAugmentation	33,604	14,096	2.384
F005	Mixes	57,753	31,077	1.858

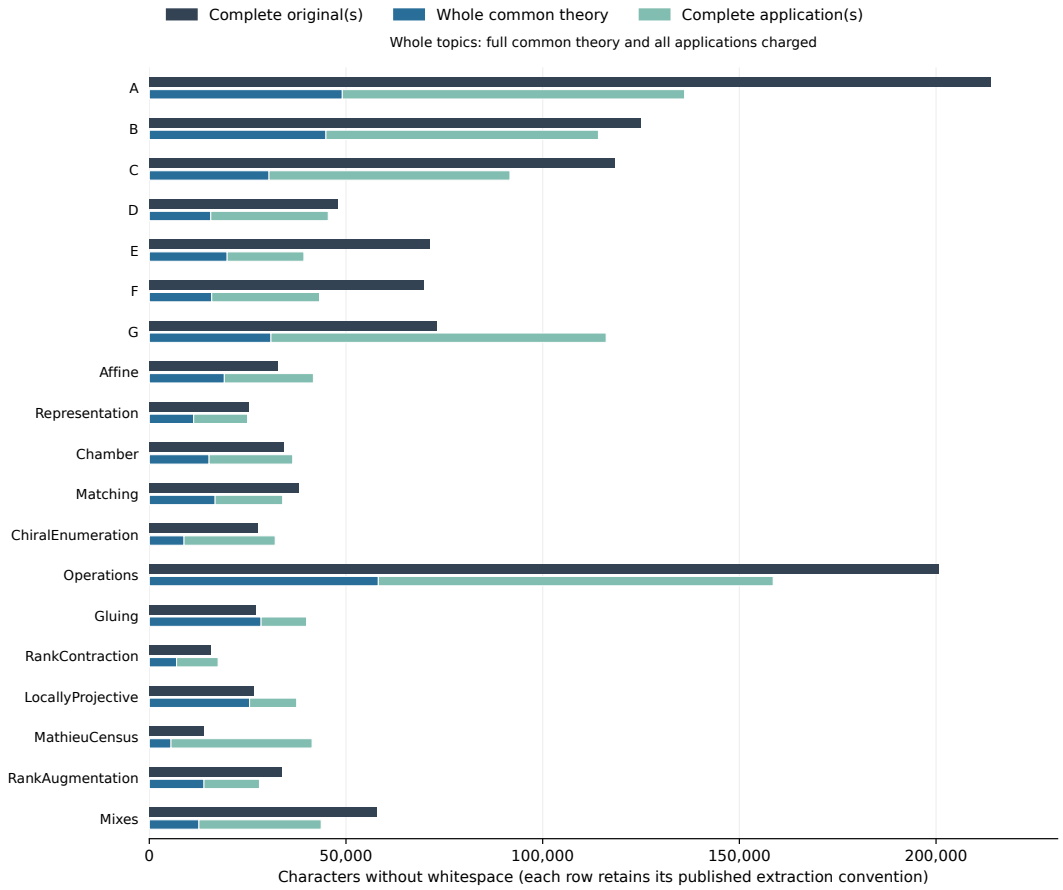
Exact exclusions. L008: three Magma functions, section 5, physical pages 8–10. L028: the ChiralPolytopes, OrderedSubsetsGroups and DiagramPlus implementations, section 4.2, physical pages 9–12. N006: the McL search pseudocode in the Theorem 5.3 proof, physical page 20. Seven blocks are removed in twelve page chunks; surrounding proof, comments outside those blocks, Lemma 4.5, mathematical algorithm prose, presentations and tables remain. No whole section is discarded.

Limits and sensitivity. The listing audit covers the measured text and supplied replacement TeX; it is not a claim about unseen software or unextracted graphics. Operations retains its proposed text proxies for source diagrams, with a separate text-only sensitivity in the matched numerical packet. That sensitivity is not substituted for the primary coordinate; graphic complexity remains unknown. The matched packet preserves all 30 coordinates per topic, plus ten text-only Operations coordinates. Unknown external proofs, source software, historical costs and complete formalization costs are null, not zero. Partial Lean checks are not full source formalization.

The prior v10 and v11 costs, factors and held statuses remain historical records. This edition supplies current graphs without changing scientific admissions or adding source, question or formalization credit.

Whole topics

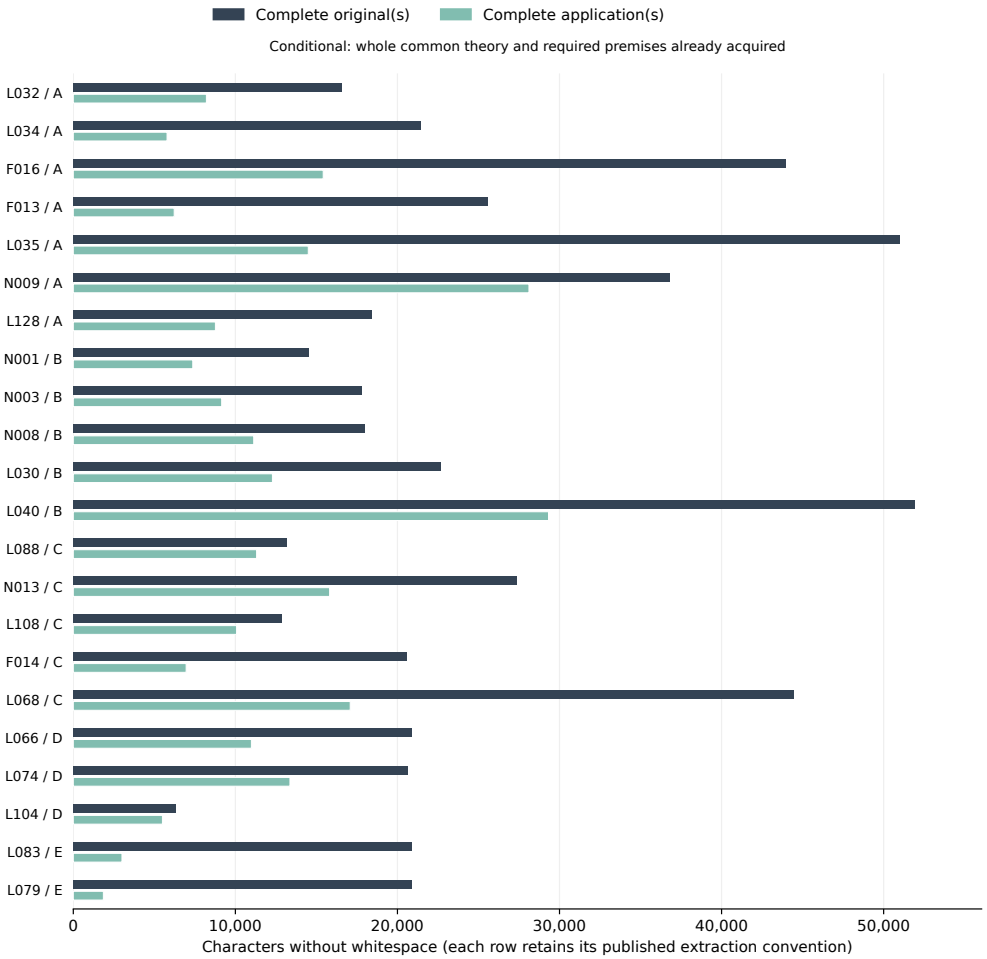
Grey bars show complete original manuscript text under each row's declared convention. Replacement bars stack whole common theory and all applications. The axis starts at zero. These are lengths; ratios are shown in the tables.



The entire common contribution remains visible, including theory needed only by some applications. The 57 actual bars correspond to all 19 topics.

Conditional lengths: first 22 originals

The common theory and required premises are acquired in advance. Grey and teal bars compare complete source and local application text; they do not remove that prerequisite cost from the whole-topic table.



Conditional lengths: remaining 21 originals

The acquisition condition is identical for every row. There is no cross-topic discount and no inferred sawtooth or node-count factor.

